MIL-A-50687(AR)
AMENDMENT 5
30 January 1981
SUPERSEDING
AMENDMENT 4
26 October 1979

MILITARY SPECIFICATION

ANTENNA ASSEMBLY AS-2049/VPS-2: 10548388
* ANTENNA ASSEMBLY AS-2049A/VPS-2: 10556312

This amendment forms a part of Military Specification MIL-A-50687(MU), dated 15 April 1971.

PAGE 1

* Paragraph 1.1: Delete in its entirety and substitute new Paragraph 1.1.

"1.1 This Specification covers an Antenna Assembly AS-2049/VPS-2 and Antenna Assembly AS-2049A/VPS-2 which are used as the antenna for the Radar Set AN/VPS-2."

* Paragraph 2, Applicable Documents, Specifications: Add the following to the list of Military Specifications:

"MIL-R-48427 Reflector and Feed Assembly: 10548028 and 10556313"

PAGE 2

* Paragraph 2, Applicable Documents, Drawings: Delete in its entirety and substitute the following:

"DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548388 Antenna Assembly, AS-2049/VPS-2

F10556312 Antenna Assembly, AS-2049A/VPS-2

B10548193 Servo Electronics

FSC 1285

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PAGE 2

C10549134 Connector, Plug, Miniature

F10549103 Base Support Assembly"

* Paragraph 2, Applicable Documents, Packaging Data Sheet: Delete in its entirety and substitute the following:

"PACKAGING DATA SHEETS

10548388 Packaging of Antenna Assembly, AS-2049/VPS-2

10556312 Packaging of Antenna Assembly, AS-2049A/VPS-2"

PAGE 3

* Paragraph 3.1: Delete Paragraph in its entirety and substitute the following new Paragraph 3.1:

"3.1 Fabrication. The Antenna Assembly shall be manufactured in accordance with drawing F10548388 or F10556312 and drawings pertaining thereto, per requirement. The subassemblies of the antenna shall have been inspected and accepted in accordance with their respective drawings or specifications. The following subassemblies are covered by the indicated specifications:

UNIT NO.	NAME	SPECIFICATION
A1	Reflector & Feed Assembly: 10548028	MIL-R-48427
A1	Reflector & Feed Assembly: 10556313	MIL-R-48427
A2	Servo Drive, Elevation: 10548062	MIL-S-50667
A3	Servo Drive, Traverse: 10548052	MIL-S-50666

PAGE 4

Paragraph 3.3.3.1 Add:

"NOTE: The requirements of 3.5.1, 3.5.2, 3.5.3 and 3.5.4 need not be checked during the exposure period."

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PAGE 4

Paragraph 3.3.4.1: Delete in its entirety and substitute new paragraph 3.3.4.1:

"3.3.4.1 Operating.- The antenna assembly shall meet the requirements of 3.4, 3.5, 3.5.5, 3.5.6 and 3.5.6.1 when stabilized thermally at any temperature between -40°F and +155°F."

Paragraph 3.5: Add:

"The antenna assembly shall be tested between the frequency range of 9.15 and 9.25 GHZ."

PAGE 5

Paragraph 3.5.1: Delete in its entirety and substitute new paragraph 3.5.1:

"3.5.1 Antenna Pattern (without radome).- The energy of any side lobe shall be at least 22 decibels (db) below the main lobe energy."

Add new paragraph as follows:

"3.5.1.1 Antenna Pattern (with radome).- The energy of any side lobe shall be at least 22 decibels (db) below the main lobe energy for the E-plane and not less than 21 decibels for the H-plane."

Paragraph 3.5.2: Delete "4.0°" and substitute "4.3°"

Paragraph 3.5.3: Delete in its entirety and substitute new paragraph 3.5.3:

"3.5.3 Gain.- The antenna assembly shall have a gain of at least 31 db without radome and at least 30 db with radome over an isotropic radiator."

Paragraph 3.5.5: Delete "1.4:1" and substitute "1.5:1"

Paragraph 3.5.6.1, Line 6, 11 and 12: Change From: "30mv" To: "300 mv"

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Paragraph 3.5.6.1c, end of paragraph: Add: "(λ^2E only)".

Paragraph 3.5.7.1: Delete in its entirety and substitute new paragraph 3.5.7.1:

"3.5.7.1 Elevation.- With the +15 volts dc applied to J1-g, J1-f, J1-E, J1-F, J1-M and J1-N of the elevation drive, the potentiometer outputs of λ^2E at J1-S, λ^3E at J1-P and λE at J1-D shall provide the following voltages using J1-h as dc return.

a. A null of not more than 2.50 volts at 0° elevation lead angle (trunnion output) for the λE potentiometer. A null of not more than 120 millivolts at 0° elevation lead angle (trunnion output) for the λ^2E and λ^3E potentiometer output.

b. Measure the following voltage outputs at ±10° elevation lead angle (corresponds to 12.25° of synchro input information).

SIGNAL	VOLTAGE
λ E	7.40 volts maximum
λ^2E	2.30 ± 0.4 volts
λ^3E	.90 ± 0.3 volts"

PAGE 6 & 7

Figures 1a, 1b, 1c, (CHART RECORDING DISPLAY) Channel 2: Change the vertical scale mv From: "20, 40, 60" To: "200, 400, 600"

PAGE 15

Table IV Item 20: Change From: "5mv/cm" To: "5 mv/mm"

PAGE 16

Figure 2 (SERVO CONTROL UNIT) Change drawing according to new Figure 2. Change Servo Director Item Number From: "18" To: "16".

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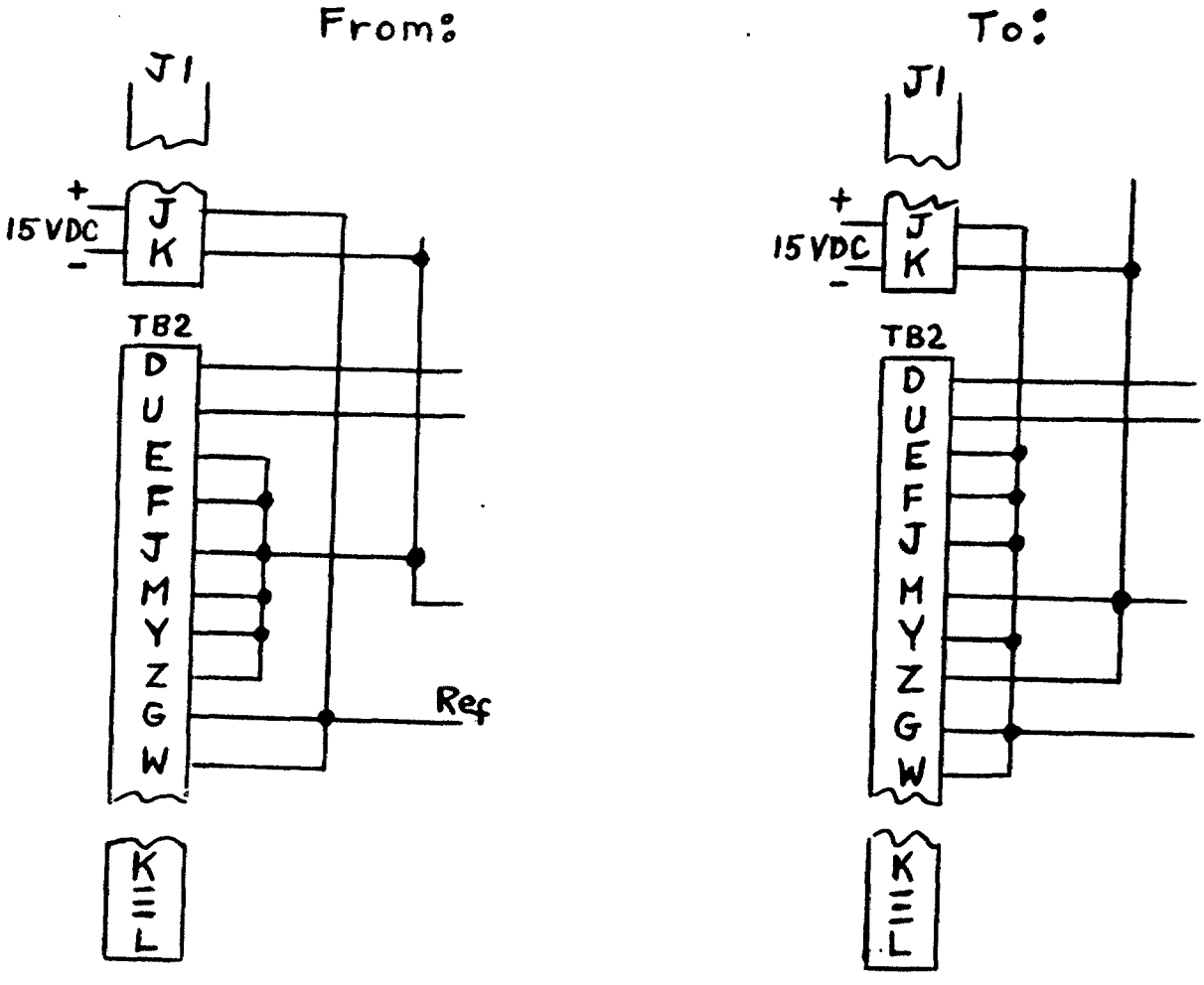


FIGURE 2
SERVO CONTROL UNIT

PAGE 19

Paragraph 4.6.1.3: Delete in its entirety and substitute new paragraph 4.6.1.3:

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"4.6.1.3 Altitude (nonoperating and operating).- This test shall be performed as specified in Procedure I, Method 500.1 of MIL-STD-810 except that in Step 2 the altitude for the operating test shall be 10,000 feet above sea level (20.6 inches HG). The operating test shall be the performance tests specified in 4.6.2, 4.6.3, 4.6.3.5, and 4.6.3.6. The antenna assembly shall also be tested to withstand an altitude of 40,000 feet, non-operating."

Paragraph 4.6.1.4.1: Delete "4.6.2 and 4.6.3 during this time" and substitute "4.6.2, 4.6.3, 4.6.3.5, and 4.6.3.6 during this time."

Paragraph 4.6.1.4.2: Delete "4.6.2 and 4.6.3" and substitute "4.6.2, 4.6.3, 4.6.3.5, and 4.6.3.6."

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* Paragraph 4.6.3: Delete the Paragraph in its entirety and substitute the following new Paragraph 4.6.3:

"4.6.3 Performance. The selected Antenna Assemblies shall be tested as follows: The tests for gain, boresight, VSWR, servo drive, dynamic test, and function potentiometers (4.6.3.3, 4.6.3.4, 4.6.3.5, 4.6.3.6, and 4.6.3.7) shall be performed on all Antenna Assemblies produced. The tests for antenna pattern (4.6.3.1) and beamwidth (4.6.3.2) need only be performed on the first article and at least 10 percent of production equipment. The tests for gain, boresight, antenna pattern, and beamwidth (4.6.3.1, 4.6.3.2, 4.6.3.3, and 4.6.3.4) may be performed on the reflector and feed subassembly, (Drawing 10548028 or Drawing 10556313) per requirement, at the discretion of the contractor."

PAGE 21

Paragraph 4.6.3.2e, g & i: Delete: "4.0°" and substitute: "4.3°"

PAGE 23

Paragraph 4.6.3.5h: Delete: "1.4:1" and substitute: "1.5:1"

Paragraph 4.6.3.6, Item d, Line 3, change From: "20 millivolt per centimeter (20 mV/cm)" To: "20 millivolt per millimeter (20 mV/mm)".

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Paragraph 4.6.3.6, Item g, Line 3, change From: "30 mV" To: "300 mV".

In Line 1 of NOTE, change From: "30 mV" To: "300 mV".

Paragraph 4.6.3.7.1: Delete in its entirety and substitute new paragraph 4.6.3.7.1:

"4.6.3.7.1 Elevation.- Using the voltmeter, Item 14 at TP1, TP2 and TP3 on the servo control unit, measurements of the λE, $\lambda^2 E$ and $\lambda^3 E$ potentiometer outputs is as specified in 3.5.7.1. Use TP5 on the servo control unit as reference for voltage measurements. When the lead angle is specified as $\pm 10^{o}$, position the dial of the servo control unit to $+12.25^{o}$."

PAGE 28

* Paragraph 5.1: Delete Paragraph in its entirety and substitute the following new paragraph 5.1:

"5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the packaging data sheet 10548388 or packaging data sheet 10556312 per requirement. The level of protection shall be as specified in the procurement document."

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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